Annex B: Decision Rubric

Challenge 2: The Quantum Routing Brain

When is quantum worth the round trip? A decision framework for your routing engine, grounded in the 2026 state of quantum hardware.

THE DEFAULT

The honest 2026 default is: always run classical.

As of April 2026, no peer-reviewed paper demonstrates unambiguous wall-clock quantum advantage on any operationally relevant defense problem. The strongest recent "utility" claims (IBM Eagle 127-qubit Ising simulation, D-Wave Advantage2 spin-glass) have each been classically dequantized within months. Your routing engine's primary job for the next 3–5 years is to keep workloads on classical accelerators while exposing quantum as a pluggable, optional backend for narrow research candidates.

The single exception is QRNG (quantum random number generation) for cryptographic keying material, which is genuinely operational today (Quantinuum Quantum Origin, NIST SP 800-90B validated April 2025).

This is not a reason to skip the quantum path. It is the reason the orchestration layer is valuable. The routing engine that gets this decision right, transparently and auditably, is the product.

THE FIVE GATES

A task should be routed to a remote quantum backend only when it passes all five gates in sequence. Failure at any gate sends the task to the classical path. Your router should implement these as sequential checks and log which gate each task passes or fails.

Gate 1: Problem Class

Does the task map cleanly to a known quantum formulation?

Problem Class	Quantum Formulation	Status (2026)
Assignment / scheduling	QUBO via QAOA or annealing	Research demos at small scale (5–13 variables). Classical ILP/Gurobi dominates at operational scale.
Graph coloring (FAP)	QUBO via QAOA / RQAOA	Dequantized for regular bipartite graphs (Bravyi et al., Quantum 6:678, 2022).
MaxCut	QAOA	Classical Goemans-Williamson SDP gives 0.878-approximation. QAOA p=1 typically reaches ~0.7 at small scale.
VRP / TSP	QUBO via QAOA or annealing	Demos at 5–13 nodes. Concorde/LKH-3 solve 100,000+ nodes classically.
Kernel classification (SVM)	Quantum kernel / QSVM	Bowles et al. (2024): classical baselines beat quantum on 160 datasets. Small-data edge cases may exist.
Linear systems	HHL algorithm	Dequantized by Tang (2018). FT-required. QRAM assumption invalid for most inputs.
Random number generation	QRNG hardware	Operational today. NIST validated. The only "route to quantum" that is unambiguously correct.

> **If the task does not appear in this table**, the answer is almost certainly “run classical.” General-purpose neural network inference, signal processing (FFT, filtering), Kalman filtering, hashing, and encryption are all definitively classical.

Gate 2: Classification Level

Can the task data leave the tactical enclave?

Level	DISA IL Equiv.	Quantum Offload?
UNCLASS	IL2	Yes. Any commercial QPU.
CUI	IL4 / IL5	Conditional. Must use an IL5-authorized cloud (Google Cloud was first hyperscaler authorized in 2022). AWS Braket and Azure Quantum operate at IL5 for some configurations. Verify before routing.
SECRET	IL6 (SIPRNet)	No. No commercial QPU is authorized at IL6. The task must remain within the classified enclave.
TS-SCI	Above IL6	No. Not offloadable under any current accreditation.

This gate is a hard constraint, not a preference. Your router must enforce it unconditionally. The stretch goal (Data Guard) addresses partial mitigation: decompose the task, strip classification-sensitive context from the quantum-eligible sub-problem, offload the sanitized sub-problem, and reintegrate. But even the Data Guard cannot offload SECRET data to a commercial QPU.

Gate 3: Latency Budget

Can the task tolerate the round trip?

The round trip to a quantum backend includes: network transit to the QPU provider (see DDIL reference, Annex D), queue time at the QPU (IBM Open Plan: minutes to hours; pay-as-you-go: seconds to minutes), circuit compilation and execution, result return, and any classical post-processing.

Deadline Class	Typical Budget	Quantum Feasible?
Hard-realtime	1–50 ms	Almost never. QPU queue times alone exceed this budget. Exception: local QRNG hardware with sub-ms latency.
Soft-realtime	50–5,000 ms	Rarely. Only if the QPU is on a low-latency link (LEO SATCOM or tactical 5G backhaul) with minimal queue, and the task is small enough for fast execution.
Batch	Seconds to hours	Yes. This is where quantum offload is architecturally feasible. Most scheduling, VRP, and mission planning tasks fall here.

Gate 4: Instance Size

Is the problem small enough for current hardware but large enough to justify the overhead?

This is the narrowest window and the reason most tasks fail the quantum path in 2026:

- **Too small:** Problems with fewer than ∼10 binary variables are solved in microseconds by brute force. The overhead of circuit compilation, QPU queue, and network transit makes quantum strictly worse.
- **Too large:** Current gate-model QPUs support ∼30 logical qubits (Heron r2: 156 physical qubits, but noise limits useful circuit depth). D-Wave hybrid solvers handle ∼5,000 binary variables but are annealers, not gate-model. Problems exceeding these limits cannot be run.
- **The window:** Roughly 15–30 qubits for gate-model, 100–5,000 variables for annealing. This window covers toy demonstrations but not operational-scale defense problems (a real FAP has thousands of variables; a real VRP has hundreds of nodes).

> **The crossover problem:** The fault-tolerant crossover for the strongest known QAOA scaling advantage (Shaydulin et al., Sci. Adv. 2024, on the LABS problem) requires ∼73.9 million physical qubits and 15 hours of runtime (Omanakuttan et al., 2025). This is not NISQ. Your router should be honest about this timeline.

Gate 5: Backend Availability

Is a quantum backend actually reachable right now?

- Is the network link to the QPU provider up? (DDIL may have severed it.)
- What is the current queue depth? (IBM Open Plan free tier: minutes to hours. Pay-as-you-go at $96/min: shorter but expensive.)
- Is the QPU calibrated and online? (Hardware goes down for recalibration regularly.)
- Does a fallback exist? (If the QPU is unavailable, the router must seamlessly fall back to classical. This fallback must be tested and reliable.)

Your router should check backend health before every offload decision, not assume availability.

THE DECISION FLOWCHART

Implement this as sequential logic in your router:

1. **Is `quantum_candidate` = Y or maybe?** No → classical.
2. **Is `classification_level` ≤ CUI?** No → classical (hard constraint).
3. **Is `latency_budget_ms` > estimated round-trip time?** No → classical.
4. **Is instance size within the QPU's feasible range?** No → classical.
5. **Is a QPU backend reachable and queue acceptable?** No → classical.
6. **All gates passed** → offload to quantum. Run classical baseline in parallel. Log both results. Compare.

> **Always run the classical baseline.** Even when all five gates pass, the quantum result should be validated against a classical solution. In 2026, the quantum path is a research instrument, not a production replacement. Your router should log the comparison and flag cases where quantum and classical disagree.

QUANTUM ADVANTAGE BY PROBLEM CLASS: THE HONEST VERDICT

Problem Class	NISQ?	FT?	2026 Recommendation
MaxCut / graph optimization	No	Marginal	Use Goemans-Williamson SDP or Gurobi. QAOA is a demo, not a solution.
TSP / VRP	No	Marginal	Use Concorde, LKH-3, or OR-Tools. Quantum demos top out at 13 nodes.
Scheduling / knapsack	No	Marginal	Use CPLEX or Gurobi MIP.
Quantum kernel ML (QSVM)	No	Unclear	Use RBF-SVM, gradient boosted trees, or neural nets. Bowles et al. (2024) found classical wins on 160/160 datasets.
QNN / variational QML	No	Unclear	Cerezo et al. (2025): all known barren-plateau-free architectures are classically simulable. Use classical deep learning.
Grover search	No (proven)	Yes	Chen et al. (2023): NISQ devices provably cannot achieve $\sqrt{N}$ speedup.
QAOA on hardware	Research	Maybe	Pilot and benchmark only. Not production.
VQE / chemistry	No	Yes	Use DMRG, CCSD(T), or DFT. Quantum relevant at $\sim$50–100 logical qubits.
HHL / linear systems	No	Yes	Dequantized by Tang (2018). Classical iterative solvers win.
QRNG	**Yes**	n/a	**Operational today.** Quantinuum Quantum Origin. Deploy.

KEY CITATIONS

- **Bowles, Ahmed, Schuld (Xanadu, 2024).** 12 QML models on 160 datasets. Classical baselines won nearly every case. Removing entanglement often improved quantum classifier performance. arXiv:2403.07059.
- **Cerezo et al. (Nat. Commun. 16:7907, 2025).** Strong evidence that all currently known barren-plateau-free variational architectures are classically simulable.
- **Chen, Cotler, Huang, Li (Nat. Commun. 14:6001, 2023).** Proved NISQ devices cannot achieve Grover-like $\sqrt{N}$ speedup.
- **Bravyi, Kliesch, Koenig, Tang (Quantum 6:678, 2022).** Level-1 QAOA cannot beat random assignment on regular bipartite graphs.
- **Shaydulin et al. (Sci. Adv. 10:eadm6761, 2024).** Best known QAOA scaling advantage (LABS problem, 1.21^N quantum vs. 1.25^N classical). But fault-tolerant crossover requires $\sim$74M physical qubits (Omanakuttan et al., 2025).
- **Tindall (PRX Quantum 5:010308, 2024).** Classical dequantization of IBM Kim et al. "utility before fault tolerance" claim.
- **Tindall (arXiv:2503.05693, 2025) and Mauron-Carleo (arXiv:2503.08247, 2025).** Contested D-Wave King et al. "beyond-classical" spin-glass claim within days of publication.
- **Quantinuum Quantum Origin.** QRNG validated under NIST SP 800-90B, April 2025. The only operationally deployed quantum technology relevant to defense compute.